

DAY 3

Discounted Cash Flow (DCF) Valuation

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

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This chapter builds working competence in Discounted Cash Flow (DCF) Valuation: the concepts, the standard procedure, and the judgment to apply it under exam conditions.

Concepts and vocabulary

Concepts and Vocabulary

Discounted Cash Flow (DCF) valuation rests on a single, powerful idea: a dollar received in the future is worth less than a dollar in hand today. Understanding why — and how to measure exactly how much less — requires mastering a small set of interlocking definitions before any formula is applied.

Core Definitions

Cash flow (CF) is the actual movement of money into or out of a business or project. Unlike accounting profit, cash flow ignores non-cash charges and focuses on real liquidity — the raw material of every DCF model.

Future Value (FV) is the nominal amount of a cash flow expected at a specific point in time. It has not yet been adjusted for the delay in receiving it.

Present Value (PV) is what that future amount is worth today, after accounting for the time value of money. Discounting converts FV into PV.

Discount rate (r) is the rate used to shrink future cash flows back to today's dollars. It reflects two things simultaneously: the opportunity cost of capital (what you could earn elsewhere) and the risk that the cash flow may not arrive as promised. In corporate practice this is often the Weighted Average Cost of Capital (WACC).

Discounted Present Value (DPV) for a single cash flow arriving n years from now is:

$$DPV = FV / (1 + r)^n$$

Net Present Value (NPV) extends this to an entire stream of cash flows — both inflows and outflows — by summing every individual DPV:

$$NPV = \sum_{t=1}^T \frac{CF_t}{(1 + r)^t} - \frac{I_0}{(1 + r)^0}$$

where t runs from period 1 to the final period of the analysis. A positive NPV signals that the investment creates value above the required return; a negative NPV destroys it.

Where DCF Fits in Practice

DCF valuation is classified as an income approach: value is derived from what an asset produces, not what was paid for it (book value) or what comparable assets trade for (market approach). Analysts at investment banks, private equity firms, and corporate finance departments use DCF as a primary or cross-check valuation tool for acquisitions, capital budgeting, and strategic planning. Because it forces explicit assumptions about growth and risk, it is especially useful when comparable market data are scarce — for example, with private companies or novel projects.

Notation Reference

| Symbol | Meaning |

|-----|-----|

| CF_t | Cash flow in period t |

| FV | Future (nominal) value of a cash flow |

| PV / DPV | Present / Discounted present value |

| r | Discount rate (per period) |

| n or t | Number of periods until cash flow occurs |

| NPV | Net present value (sum of all DPVs) |

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Worked Example: Terminology Applied to a Mini Scenario

Scenario. A small equipment rental business expects to receive \$15,000 at the end of Year 1, \$18,000 at the end of Year 2, and \$20,000 at the end of Year 3. The appropriate discount rate is 10% per year. What is the NPV of this cash flow stream, ignoring any initial investment for now?

Step 1 — Identify each FV and its timing.

- FV₁ = \$15,000, t = 1
- FV₂ = \$18,000, t = 2
- FV₃ = \$20,000, t = 3

Step 2 — Apply the DPV formula to each cash flow.

$$\text{DPV}_1 = 15,000 / (1.10)^1 = 15,000 / 1.1000 = \$13,636.36$$

$$\text{DPV}_2 = 18,000 / (1.10)^2 = 18,000 / 1.2100 = \$14,876.03$$

$$\text{DPV}_3 = 20,000 / (1.10)^3 = 20,000 / 1.3310 = \$15,026.30$$

Step 3 — Sum the discounted values to find NPV.

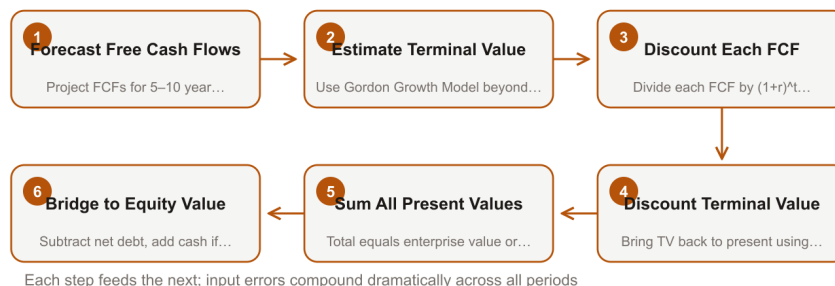
$$\text{NPV} = \$13,636.36 + \$14,876.03 + \$15,026.30 = \$43,538.69$$

Interpretation. The three future cash flows totaling \$53,000 in nominal terms are worth only \$43,538.69 today at a 10% discount rate. The \$9,461.31 difference is the cost of waiting — precisely what DCF quantifies.

Common mistake: Students often apply the discount rate to the sum of all cash flows rather than discounting each period's cash flow individually before adding. Because each cash flow sits at a different point in time, each requires its own exponent n. Combining undiscounted flows first and discounting once produces a materially wrong — and always too high — present value.

The standard procedure

DCF Valuation: Standard Procedure



Each step feeds the next; input errors compound dramatically across all periods

Standard Procedure for DCF Valuation

Discounted cash flow valuation rests on a single economic idea: a dollar received in the future is worth less than a dollar today. The procedure translates that idea into a disciplined, repeatable sequence that converts projected operating performance into an estimate of intrinsic value. Mastering each stage—inputs, steps, outputs, and checks—is essential for the exam and for real-world practice.

Inputs You Must Gather First

Before any calculation begins, assemble four categories of data:

- Projected free cash flows (FCFs) for a discrete forecast horizon, typically five to ten years.
- A discount rate that reflects the riskiness of those cash flows—usually the weighted average cost of capital (WACC) for a firm or the required return for a project.
- A terminal value assumption—either a perpetuity growth rate or an exit multiple—to capture value beyond the forecast horizon.
- The current date and cash-flow timing convention (end-of-period is standard; mid-year adjustment is used for continuous operations).

Garbage in, garbage out: errors in the discount rate or growth rate compound dramatically, so verify each input independently before proceeding.

Step-by-Step Procedure

- Forecast free cash flows for each year of the explicit period.
- Estimate terminal value (TV) at the end of year n using the Gordon Growth Model:

$$TV = FCF_{(n+1)} / (r - g)$$

where r is the discount rate and g is the long-run sustainable growth rate.

- Discount each annual FCF to present value:

$$\text{PV of FCF}_t = \text{FCF}_t / (1 + r)^t$$

- Discount the terminal value to present value:

$$\text{PV of TV} = \text{TV} / (1 + r)^n$$

- Sum all present values to obtain enterprise value (or project NPV):

$$\text{Enterprise Value} = \sum_{t=1}^n \frac{\text{FCF}_t}{(1+r)^t} + \frac{\text{TV}}{(1+r)^n}$$

- Bridge to equity value if needed: subtract net debt and add cash.

Fully Worked Numerical Example

Scenario: You are valuing a mid-sized manufacturer. The WACC is 10%, the long-run growth rate is 3%, and the explicit forecast horizon is five years.

| Year | FCF (\$M) | Discount Factor $1/(1.10)^t$ | PV (\$M) |

|-----|-----|-----|-----|

| 1 | 12.0 | 0.9091 | 10.91 |

| 2 | 13.5 | 0.8264 | 11.16 |

| 3 | 14.8 | 0.7513 | 11.12 |

| 4 | 16.0 | 0.6830 | 10.93 |

| 5 | 17.5 | 0.6209 | 10.87 |

Sum of discounted FCFs = \$54.99 M

Step 2 — Terminal value (Year 6 FCF = $17.5 \times 1.03 = \$18.025$ M):

$$\text{TV} = \frac{18.025}{0.10 - 0.03} = \frac{18.025}{0.07} = \$257.50 \text{ M}$$

Step 4 — PV of terminal value:

$$\text{PV of TV} = \frac{257.50}{(1.10)^5} = \frac{257.50}{1.6105} = \$159.89 \text{ M}$$

Step 5 — Enterprise value:

$$\text{EV} = 54.99 + 159.89 = \$214.88 \text{ M}$$

Step 6 — Equity value: Net debt is \$30 M; cash is \$5 M.

$$\text{Equity Value} = 214.88 - 30 + 5 = \$189.88 \text{ M}$$

If there are 10 million shares outstanding, the implied share price is \$18.99.

Outputs and What They Tell You

The primary output is enterprise value, which represents the total value of operations to all capital providers. The secondary output is equity value per share, the figure most directly comparable to a market price. The ratio of terminal value to total enterprise value—here about 74%—signals how sensitive the result is to your long-run assumptions.

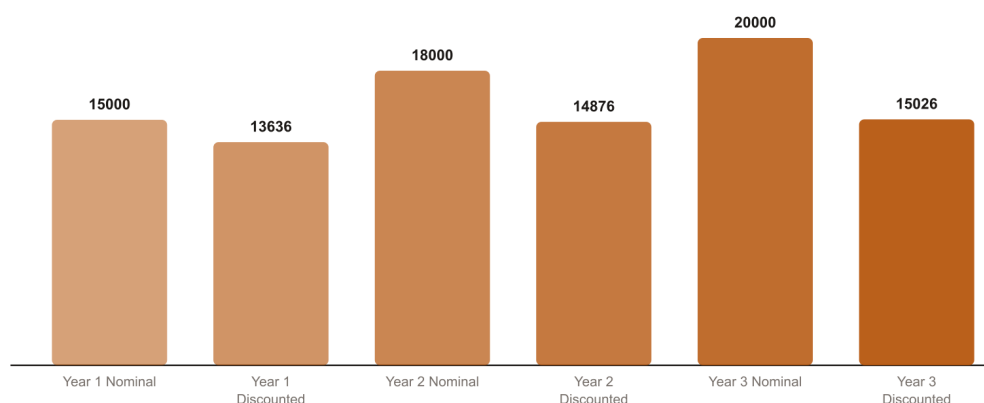
Checks to Perform

- Sensitivity analysis: Flex the discount rate $\pm 1\%$ and the growth rate $\pm 0.5\%$. If value swings wildly, your inputs need more scrutiny.
- Cross-check with multiples: Compare your implied EV/EBITDA to industry peers. A large divergence demands explanation.
- Terminal value proportion: A TV share above 80% of total value suggests the explicit forecast period may be too short or FCFs are understated.
- Sign and magnitude sanity: Confirm FCFs are positive (or intentionally negative in early years) and that the discount rate exceeds the growth rate—otherwise the Gordon formula produces a nonsensical negative or infinite value.

Common mistake: Candidates frequently forget to grow the final forecast-year FCF by $(1 + g)$ before dividing in the Gordon Growth Model. Using Year 5 FCF directly instead of Year 6 FCF understates terminal value and, in this example, would reduce EV by roughly \$11 M—a material error.

Applications and pitfalls

Nominal vs. Discounted Cash Flows (10%)



Nominal total \$53,000 vs. NPV \$43,538 — \$9,461 is the measurable cost of waiting

Nominal total \$53,000 vs. NPV \$43,538 — \$9,461 is the measurable cost of waiting

DCF valuation is one of the most powerful tools in a financial analyst's kit — and one of the most frequently misused. Understanding where practitioners go wrong is just as important as knowing the mechanics.

Typical Use Cases

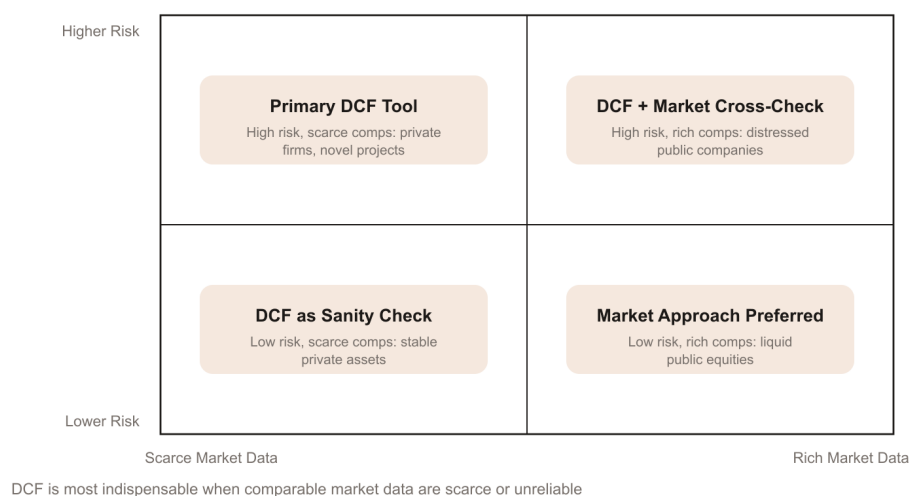
DCF analysis appears across a wide range of professional contexts:

- Equity valuation — estimating the intrinsic value of a stock by projecting free cash flows to equity or to the firm
- Capital budgeting — deciding whether a project's NPV justifies the required investment
- M&A due diligence — stress-testing an acquisition price against realistic cash flow scenarios
- Startup and venture capital valuation — discounting projected cash flows when no market comparables exist
- Real estate and infrastructure — appraising long-lived assets whose value depends almost entirely on future income streams

In each case the analyst forecasts cash flows, selects an appropriate discount rate (typically WACC for firm-level analysis), and sums the present values. The terminal value — representing cash flows beyond the explicit forecast horizon — often accounts for 60–80% of total value, which is precisely where errors concentrate.

Common Mistakes

DCF Applicability: Risk vs. Market Data



DCF is most indispensable when comparable market data are scarce or unreliable

1. Mixing nominal and real figures. Projecting cash flows in real (inflation-adjusted) terms and then discounting at a nominal rate — or vice versa — is a classic consistency error. Always match: nominal flows with a nominal rate, real flows with a real rate.
2. Double-counting tax. Analysts sometimes apply a tax shield in the cash flows and use an after-tax WACC that already embeds the shield.
3. Anchoring the terminal growth rate too high. A perpetuity growth rate above long-run GDP growth implies the company eventually becomes larger than the entire economy. Rates above 3–4% for mature firms deserve heavy scrutiny.

4. Ignoring working capital and capex. Free cash flow is not net income. Omitting changes in net working capital or treating maintenance capex as optional inflates projected cash flows.

5. Discounting from the wrong date. Assuming all cash flows arrive at year-end when they actually arrive mid-year understates value.

Common mistake: Analysts frequently use EBITDA as a proxy for free cash flow, forgetting to subtract taxes, changes in working capital, and capital expenditures. This can overstate intrinsic value by 30% or more in capital-intensive industries.

Worked Example: A Flawed Attempt, Corrected

A candidate values a small manufacturing firm. Their first attempt contains three errors. We fix each one.

Given information:

- Year 1–5 projected EBITDA: \$500,000 per year (flat)
- Depreciation: \$80,000/year; Capex: \$120,000/year; 9Eworking capital: \$20,000/year increase
- Tax rate: 25%; WACC: 10%; Terminal growth rate assumed: 6%

Step 1 — Identify the flawed free cash flow

The candidate used EBITDA directly as cash flow: FCF (wrong) = \$500,000.

Correct free cash flow:

$$\text{FCF} = \text{EBITDA} \times (1 - \text{tax rate}) - (\text{Capex} - \text{Depreciation}) - \Delta \text{WC}$$

$$\text{FCF} = \$500,000 \times 0.75 - (\$120,000 - \$80,000) - \$20,000$$

$$\text{FCF} = \$375,000 - \$40,000 - \$20,000 = \$315,000$$

Step 2 — Discount the five-year cash flows correctly

PV of FCF = $\sum_{t=1}^5 \frac{\text{FCF}_t}{(1+r)^t}$ or $n = 1$ to 5

Year	FCF	Discount factor	PV
1	\$315,000	0.9091	\$286,364
2	\$315,000	0.8264	\$260,322
3	\$315,000	0.7513	\$236,656
4	\$315,000	0.6830	\$215,145
5	\$315,000	0.6209	\$195,584
Sum of PVs = \$1,194,071			

Step 3 — Correct the terminal growth rate

The candidate used $g = 6\%$, which exceeds long-run nominal GDP growth. A defensible rate for a mature manufacturer is $g = 3\%$.

$$\text{Terminal Value (Year 5)} = \text{FCF}_{\text{Year 5}} \times (1 + g) \div (\text{WACC} - g)$$

$$\text{TV} = \$315,000 \times 1.03 \div (0.10 - 0.03) = \$324,450 \div 0.07 = \$4,635,000$$

$$\text{PV of Terminal Value} = \$4,635,000 \div (1.10)^5 = \$4,635,000 \times 0.6209 = \$2,877,872$$

Step 4 — Sum to intrinsic value

$$\text{Intrinsic Value} = \$1,194,071 + \$2,877,872 = \$4,071,943$$

The candidate's flawed model (using raw EBITDA and $g = 6\%$) produced a value of roughly \$7.9 million — nearly double the corrected figure.

Sanity Checks

Before presenting any DCF result, run these tests:

- Implied exit multiple — divide terminal value by Year 5 EBITDA. Does the implied EV/EBITDA multiple match industry comparables?
- Sensitivity table — vary WACC $\pm 1\%$ and terminal growth $\pm 0.5\%$. If small changes swing value by 50%, your assumptions need revisiting.
- Cross-check with comparables — does your intrinsic value per share sit within the range suggested by trading multiples?
- Nominal vs. real consistency check — confirm every input uses the same inflation basis.
- FCF bridge — reconcile net income to free cash flow line by line to catch omitted items.

A DCF is only as credible as its assumptions. Disciplined sanity-checking transforms a mechanical exercise into a genuinely useful valuation.

